Software Requirements Specification
Version 1.0

PlantPalace

Theme: eGreenNursery

Category: Website Design and Development

Contents

1.1 Background and Necessity for the Website

The necessity for a plant nursery Website stems from several factors related to the growing interest in gardening, increased demand for plants, and the convenience of online shopping. Here are some reasons why such a Website would be required:

- Increased Interest in Gardening:

  There has been a significant rise in the popularity of gardening and plant ownership, driven by various factors such as desire for a green environment, stress relief, and aesthetic appeal. As more people engage in gardening as a hobby or lifestyle, the necessity for easily accessible and diverse plant options becomes crucial.

- Wide Variety of Plants:

  Plant nurseries offer a wide variety of plant species, each with unique characteristics, care requirements, and purposes. A dedicated Website allows the nursery to showcase their extensive collection and provide detailed information to potential customers, enabling them to make informed decisions based on their specific requirements and preferences.

- Convenience and Accessibility:

  An e-commerce as well as informational Website provides convenience by allowing customers to browse and purchase plants from the comfort of their homes. Users can access the Website at any time, eliminating the constraints of physical store hours and geographical limitations. This accessibility increases the reach of the nursery, attracting customers from different locations.

- Plant Information:

  A plant nursery e-commerce Website serves as an informational platform for customers. Providing detailed information about each plant species, including origin, care requirements, and watering cycle, helps customers understand the plants they are interested in.

This information promotes responsible gardening practices and ensures that customers select plants suitable for their environment and abilities.

❖ Search, Sorting, and Filtering Capabilities:

With a wide range of plant options, users may have specific preferences such as indoor plants, succulents, or flowering shrubs. The Website's search, sorting, and filtering features allow users to find plants that match their criteria efficiently. This saves time and enhances the overall user experience by narrowing down options based on their preferences.

❖ Customer Engagement and Loyalty:

A well-designed Website can foster customer engagement and loyalty by offering features such as wishlists, personalized recommendations, and informative blog content related to gardening and plant care. This engagement strengthens the relationship between the nursery and its customers, increasing the likelihood of repeat purchases and referrals.

1.2 Proposed Solution

The proposed solution is a Website called 'PlantPalace' for a plant nursery. The Website should serve as an e-commerce as well as informational platform where users can browse and purchase various plants. It should provide categories for different types of plants, allow users to search for specific plants, and offer sorting and filtering options. Additionally, detailed information about each plant species should be displayed, including origin country and watering cycle.

1.3 Purpose of the Document

The purpose of this document is to present a detailed description of the Website, **PlantPalace,** an online plant nursery through which users can shop for plants, gain information on plant varieties and species, browse through plant catalogs, and so on.

This document explains the purpose and features of the Website, the interfaces of the Website, what the Website will do, and the constraints under which it must operate. This document is intended for both stakeholders and developers of the Website.

1.4 Scope of Project

This Web portal will be a responsive and visually appealing Website to be used by individuals.

1.5 Constraints

The Web portal will not have any facility to store information on the server. Information can be fetched from JSON/TXT files and users can view the same being displayed, however, information cannot be written to the files from within the portal.

1.6 Functional Requirements

The portal will be designed as a Single-Page-Application (SPA) and responsive Website with a set of pages and menus that represent choice of activities to be performed. The pages, menus, and other visual elements must be designed in a visually appealing manner with attractive fonts, colors, and animations. All of these should also be laid out in a responsive manner.

Following are the functional requirements of the portal:

Catalog Display:

The Website should display a catalog of plants available for purchase. Plants should be organized into categories, such as indoor, outdoor, succulents, flowering shrubs, and so on. Users should be able to browse through the catalog and view individual plant listings.

Data for this can be hardcoded or retrieved from the JSON/TXT file.

Search and Navigation:
Users should be able to search for a specific plant by name or keywords. The Website should provide navigation options to access different plant categories and sections.

Sorting and Filtering:
Users should be able to sort the plant catalog based on various criteria, such as price, popularity, or alphabetical order.

Users should be able to filter plants based on specific attributes, such as indoor/outdoor, light requirements, water requirements, and so on.

Plant Information:
Each plant listing should display detailed information about the plant species. Information should include the plant's common and scientific name, origin country, watering cycle, light requirements, and other relevant details.

Data for this can be hardcoded or retrieved from the JSON/TXT file.

Shopping Cart and Checkout:
Users should be able to add plants to their shopping cart for purchase. The Website should display the contents of the user's shopping cart and allow them to modify quantities or remove items.

Feedback:
Feedback menu option should enable users to provide their feedback about this Website through a feedback form.

Contact Us:
Contact Us menu option should enable users to contact the creators of the Website. An email id can be displayed here for contact information.

Sitemap: To help users understand the flow of the Website, you will create a Sitemap and add it to the home page of your Website.

Shopping cart checkout or payment functionality are not required to be implemented.

Technical Considerations:

- Use HTML5, CSS3, Figma Toolkit, Bootstrap, jQuery, and JavaScript to build the User Interface (UI).
- Use either Angular or ReactJS to develop the application's frontend and add dynamic and responsive features along with SPA functionality.
- Use JSON/TXT files to handle data retrieval.
- Ensure the application is responsive and compatible with different screen sizes.

Tasks:

1. Design the UI for the PlantPalace Website, including the catalog display, search bar, sorting options, and plant information section.
2. Create necessary components and services to display the plant catalog, handle search, sorting, and filtering operations.
3. Implement the shopping cart (adding/modifying/deleting item) basic functionalities.
4. Optionally, use REST APIs if you are well-versed with them.
5. Test the Website's functionality, including browsing, searching, adding items to the cart, and placing orders. Deploy the Website to a local Web server such as XAMPP for testing purposes.

Important: Do NOT use boilerplate templates or readymade templates for development as it will adversely affect your evaluation. Your own Website design and development skills will be tested, hence no third party templates should be used here.

1.7 Non-Functional Requirements

There are several non-functional requirements that should be fulfilled by the Website.

The Website should be:

Safe to use: The Website should not result in any malicious downloads or unnecessary file downloads.

Accessible: The Website should have clear and legible fonts, user-interface elements, and navigation elements.

User-friendly: The Website should be easy to navigate with clear items and other elements and easy to understand.

Operability: The Website should operate in a reliably efficient manner.

Performance: The Website should demonstrate high value of performance through speed and throughput. In simple terms, the Website should be fast to load and page redirection should be smooth.

Capacity: The Website should support large number of users.

Availability: The Website should be available 24/7 with minimum downtime.

Compatibility: The Website should be compatible with latest browsers.

These are the bare minimum expectations from the project. It is a must to implement the functional and non-functional requirements given in this SRS. Once they are complete, you can use your own creativity and imagination to add more features if required.

1.8 Interface Requirements

1.8.1 Hardware

Intel Core i5 Processor or higher
8 GB RAM or higher
Color SVGA
500 GB Hard Disk space
Mouse
Keyboard

1.8.2 Software

Technologies to be used:

1. Frontend: HTML5, CSS3, Bootstrap (optional), JavaScript, Figma Toolkit, jQuery, AngularJS/Angular 9/ReactJS, and XML
2. Data Store: JSON/TXT

1.9 Project Deliverables

You will design and build the project and submit it along with a complete project report that includes:

- Problem Definition
- Design specifications
- Diagrams such as flowcharts for various activities, Data Flow Diagrams, and so on
- Source Code
- Test Data Used in the Project
- Project Installation Instructions (if any)

Documentation is considered as a very important part of the project. Ensure that documentation is complete and comprehensive. The consolidated project will be submitted as a zip file with a ReadMe.doc file listing assumptions (if any) made.

Submit a video clip demonstrating the working of the Website. Optionally, a live hosted URL can be supplied for the site. Over and above the given specifications, you can apply your creativity and logic to improve the portal.

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